

The Money Supply Process

Chapter 14

Econ 351 – Monetary Economics

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Preview

Movements in the money supply affect interest rates, inflation, and economic activity, making it crucial to understand how the money supply is determined.

- ▶ Who controls it?
- ▶ What causes it to change?
- ▶ How can its control be improved?

This chapter explains the **money supply process**, focusing on how banks create deposits—the largest component of the money supply.

Outline

Three Players in the Money Supply Process

The Fed's Balance Sheet

Control of the Monetary Base

Multiple Deposit Creation

The Money Multiplier

Application: Quantitative Easing

Three Players in the Money Supply Process

1. The Central Bank

The government agency that oversees the banking system and is responsible for the conduct of monetary policy.

In the United States, this is the Federal Reserve System (the Fed).

2. Banks (Depository Institutions)

Financial intermediaries that accept deposits from individuals and institutions and make loans.

Includes: commercial banks, savings and loan associations, mutual savings banks, and credit unions.

3. Depositors

Individuals and institutions that hold deposits in banks.

Of the three players, the **central bank**—the Federal Reserve System—is the most important. The Fed's conduct of monetary policy involves actions that affect its balance sheet (holdings of assets and liabilities), which we will now examine.

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The Fed's Balance Sheet

Federal Reserve System	
Assets	Liabilities
Securities	Currency in circulation
Loans to financial institutions	Reserves

Simplified balance sheet of the Federal Reserve System.

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Key Focus: Changes in assets/liabilities directly impact the **monetary base** and money supply.

The Fed's Liabilities

▶ **Currency in circulation**

- ▶ Physical cash held by the public (Fed-issued notes)
- ▶ Example: \$100 bill = Fed's liability, exchangeable only for other Fed notes

▶ **Reserves**

- ▶ Deposits at the Fed + Bank vault cash
- ▶ **Required reserves:** Mandated by Fed (e.g., 10% of deposits)
- ▶ **Excess reserves:** Voluntarily held by banks

Monetary Base = Fed's liabilities = currency in circulation + reserves

The Fed's Assets: Drivers of Monetary Policy

1. Securities

- ▶ Primarily U.S. Treasury bonds
- ▶ Fed buys/sells securities to inject/absorb reserves (*open market operations*)

2. Loans to financial institutions

- ▶ Called **discount loans** (or "borrowed reserves")
- ▶ Interest rate charged = *discount rate*

Why Assets Matter:

- ▶ Asset changes directly alter reserves/monetary base
- ▶ Fed earns income from assets (e.g., interest on securities)

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Control of the Monetary Base

The monetary base (**MB**) consists of:

$$MB = C + R$$

- ▶ C = Currency in circulation (Federal Reserve & Treasury notes)
- ▶ R = Total reserves in banking system (deposits at Fed + vault cash)

Fed's Tools for Control:

- ▶ Open market operations (primary tool)
- ▶ Loans to financial institutions

Open Market Operations

Key Mechanism

Open Market Purchase	Open Market Sale
$\uparrow \text{ Reserves} \Rightarrow \uparrow \text{ MB}$	$\downarrow \text{ Reserves} \Rightarrow \downarrow \text{ MB}$

Example: \$100M Bond Purchase

Fed's Balance Sheet

Assets	Liabilities
Securities +\$100M	Reserves +\$100M

Banking System

Assets	Liabilities
Securities -\$100M	
Reserves +\$100M	

Key Insight: Open market operations change MB 1:1 with transaction size.

Other Dynamics Affecting MB

▶ **Deposits → Currency Shifts**

Example: \$100M cash withdrawal:

- ▶ Public: ↑ Currency \$100M, ↓ Deposits \$100M
 - ▶ Banks: ↓ Reserves \$100M
 - ▶ Fed: ↑ Currency \$100M, ↓ Reserves \$100M
- ⇒ **No net change to MB**

▶ **Loans to Financial Institutions**

Example: \$100M Fed loan:

- ▶ Fed: ↑ Loans \$100M, ↑ Reserves \$100M
- ⇒ ↑ MB by \$100M
- ▶ Loan repayment has opposite effect

Limits of Fed Control

- ▶ **Float**

Temporary reserve increase during check-clearing

Caused by: Time lag in check processing (e.g., weather delays)

- ▶ **Treasury Deposits at Fed**

Treasury moves funds from banks → Fed

⇒ ↓ Bank reserves ⇒ ↓ MB

- ▶ **Foreign Exchange Interventions**

Fed buys/sells foreign currency → impacts MB

Key Point: These factors cause short-term MB fluctuations but are *not* controlled by the Fed.

Fed's Control Framework

Decomposing the Monetary Base

$$MB = \underbrace{MB_n}_{\text{Controlled}} + \underbrace{BR}_{\text{Less Controlled}}$$

- ▶ **Nonborrowed Base (MB_n):** Set via open market operations
- ▶ **Borrowed Reserves (BR):** Banks' discount loans (depends on bank decisions)

Why It Matters:

- ▶ Fed fully controls MB_n but not BR
- ▶ Short-term MB volatility driven by float/Treasury actions
- ▶ Long-term MB trends dominated by open market operations

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Multiple Deposit Creation: Single Bank

Key Scenario

Fed injects \$100M reserves via open market purchase to First National Bank:

- ▶ Initial T-account: Reserves $\uparrow$ \$100M, Securities $\downarrow$ \$100M
- ▶ Bank loans out excess reserves: Loans $\uparrow$ \$100M, Deposits $\uparrow$ \$100M

Why It Stops:

- ▶ Borrowers spend deposits $\rightarrow$ Reserves flow to other banks
- ▶ Single bank's max loan = initial excess reserves (\$100M)

First National Bank: Final T-Account

Assets

Liabilities

Loans +\$100M

Deposits +\$100M

Securities $-\$100\text{M}$

Deposit Expansion in the Banking System

► Assumptions:

- 10% reserve ratio
- No excess reserves held

► Process:

- Bank A: \$100M deposit → \$90M loan
- Bank B: \$90M deposit → \$81M loan
- Repeated until reserves are exhausted

Total Deposit Creation

$$\begin{aligned}\Delta D &= \frac{1}{rr} \times \Delta R \\ &= \frac{1}{0.10} \times 100 \\ &= \$1,000\text{M}\end{aligned}$$

Bank	Deposits ↑	Loans ↑	Reserves ↑
A	\$100M	\$90M	\$10M
B	\$90M	\$81M	\$9M
C	\$81M	\$72.9M	\$8.1M
...	...	...	...
Total: \$1,000M deposits created			

Simple Deposit Multiplier

Formula

$$\Delta D = \frac{1}{rr} \times \Delta R \quad \text{where } rr = \text{required reserve ratio}$$

Derivation (banks hold no excess reserves, so $RR = R$):

1. Total reserves $R = rr \times D$
2. Rearrange: $D = \frac{1}{rr} \times R$
3. Take changes: $\Delta D = \frac{1}{rr} \times \Delta R$

Key Insight:

- ▶ Multiplier = $\frac{1}{rr}$ (e.g., 10 for a 10% reserve ratio)
- ▶ Works equivalently for loans *or* securities purchases

Single Bank vs. Banking System

Single Bank:

- ▶ Limited to initial excess reserves
- ▶ Loses reserves when deposits leave
- ▶ Max deposit creation = \$100M

First National Bank	
Assets	Liabilities
Securities	-\$100m
Reserves	+\$100m

First National Bank	
Assets	Liabilities
Securities	-\$100m
Reserves	+\$100m
Loans	+\$100m

First National Bank	
Assets	Liabilities
Securities	-\$100m
Reserves	+\$100m

T-account of a single bank after lending out \$100M of excess reserves.

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Entire Banking System:

- ▶ Reserves recirculate between banks
- ▶ Cumulative effect: \$1,000M deposits
- ▶ Full multiplier effect achieved

Bank	Increase in Deposits (\$)	Increase in Loans (\$)	Increase in Reserves (\$)
First National	0.00	100.00 m	0.00
A	100.00 m	90.00 m	10.00 m
B	90.00 m	81.00 m	9.00 m
C	81.00 m	72.90 m	8.10 m
D	72.90 m	65.61 m	7.29 m
E	65.61 m	59.05 m	6.56 m
F	59.05 m	53.14 m	5.91 m
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
Total for all banks	1,000.00 m	1,000.00 m	100.00 m

Deposit creation across the banking system (10% reserve requirement).

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Financial Markets

Critique of the Simple Model

- ▶ Holding cash stops the process: currency has no multiple deposit expansion.
- ▶ Banks may not use all of their excess reserves to buy securities or make loans.
- ▶ Depositors' decisions (how much currency to hold) and banks' decisions (how many excess reserves to hold) also cause the money supply to change.

Key idea: the Fed is not the only player. The simple multiplier $1/rr$ is an upper bound; the actual money supply depends on the choices of depositors and banks as well as the Fed.

Factors That Determine the Money Supply

Player	Variable	Change	Money supply
Fed	Nonborrowed monetary base, MB_n	↑	↑
Fed	Required reserve ratio, r	↑	↓
Banks	Borrowed reserves, BR	↑	↑
Banks	Excess reserves, ER	↑	↓
Depositors	Currency holdings, C	↑	↓

- ▶ Money supply is **positively** related to MB_n and BR .
- ▶ Money supply is **negatively** related to r , excess reserves, and currency holdings.

Note: only increases in the variables are shown; decreases have the opposite effect on the money supply.

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The Money Multiplier

Define money as currency plus checkable deposits (M1):

$$M = MB \times m$$

where m is the **money multiplier**.

Assumption: desired holdings of currency C and excess reserves ER grow proportionally with checkable deposits D :

$$C = c \times D \qquad ER = e \times D$$

- ▶ c = currency ratio (chosen by depositors)
- ▶ e = excess reserves ratio (chosen by banks)

Deriving the Money Multiplier (1/2)

Total reserves equal required reserves plus excess reserves:

$$R = RR + ER$$

Required reserves equal the required reserve ratio times checkable deposits ($r < 1$):

$$RR = r \times D \quad \Rightarrow \quad R = r \times D + ER$$

The monetary base equals currency plus reserves:

$$MB = C + R$$

This shows the amount of monetary base needed to support the existing amounts of checkable deposits, currency, and excess reserves.

Deriving the Money Multiplier (2/2)

Substituting $C = cD$, $RR = rD$, $ER = eD$:

$$MB = (c \times D) + (r \times D) + (e \times D)$$

Divide both sides by the term in parentheses:

$$D = \frac{MB}{c + r + e}$$

Substituting again:

$$M = D + C = D + (c \times D) = D \times (1 + c)$$

The money multiplier is then:

$$m = \frac{1 + c}{c + r + e}$$

Intuition Behind the Money Multiplier

Given:

$$r = 0.10, \quad C = \$400\text{B}, \quad D = \$800\text{B}, \quad ER = \$0.8\text{B}$$

$$c = \frac{400}{800} = 0.5, \quad e = \frac{0.8}{800} = 0.001$$

$$m = \frac{1 + 0.5}{0.5 + 0.10 + 0.001} = \frac{1.5}{0.601} = 2.5$$

This is less than the simple deposit multiplier ($1/r = 10$). Although there is multiple expansion of deposits, there is no such expansion for currency: every dollar the public holds as cash is a dollar of monetary base that supports no deposits.

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Quantitative Easing and the Money Supply, 2007–2020

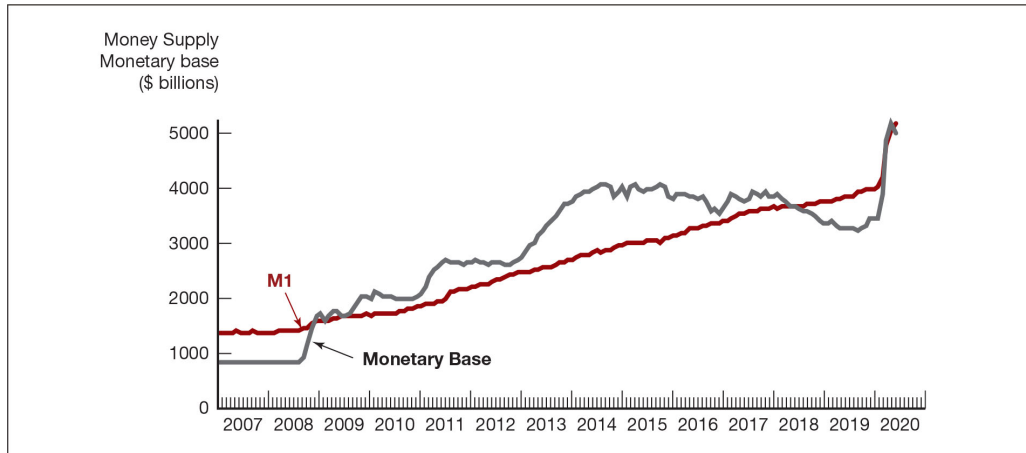
- ▶ When the global financial crisis began in the fall of 2007, the Fed initiated lending programs and large-scale asset-purchase programs – “quantitative easing” – to bolster the economy.
- ▶ By the fall of 2017 these purchases had led to a quintupling of the Fed’s balance sheet and a 350% increase in the monetary base.
- ▶ The increase in the monetary base did **not** lead to an equivalent change in the money supply because excess reserves rose dramatically.
- ▶ When the coronavirus pandemic struck in March 2020, the Fed again engaged in massive quantitative easing; the effects on the monetary base and money supply followed a similar pattern.



Ben Bernanke, Fed Chair 2006–2014, who led the first quantitative easing programs.

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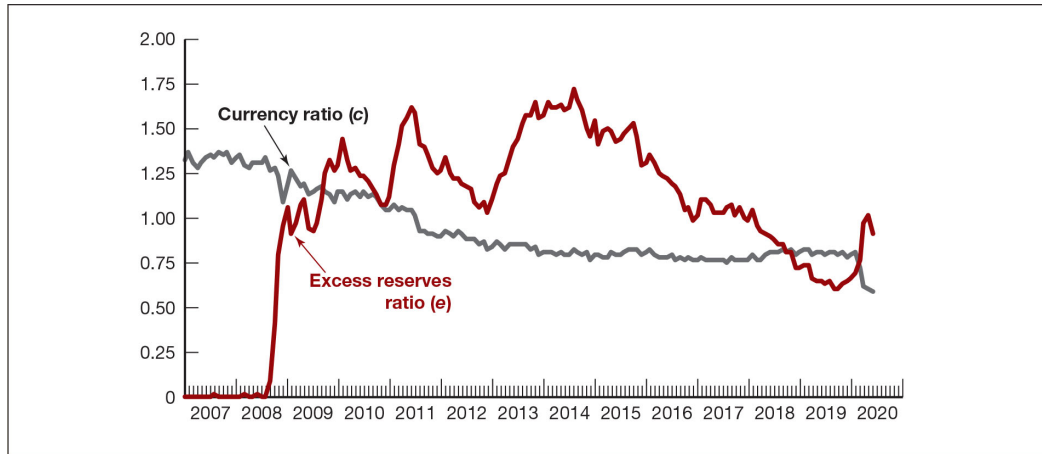
M1 and the Monetary Base, 2007–2020



The monetary base rose far more than M1 after 2008: the multiplier fell.

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Excess Reserves and Currency Ratio, 2007–2020



The excess reserves ratio e jumped from near zero after 2008 while the currency ratio c drifted down.

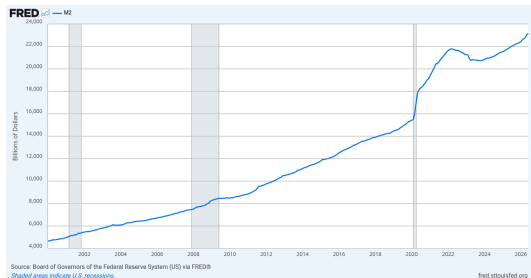
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Monetary Base vs. M2 in the Data



Monetary base (billions of dollars), 2000–2026: step jumps in 2008 and 2020 are QE.

Source: FRED, Federal Reserve Bank of St. Louis

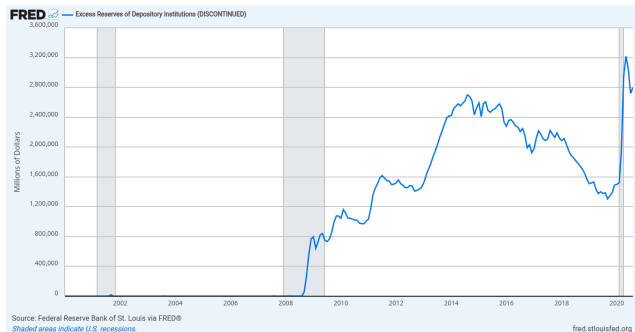


M2 money stock (billions of dollars), 2000–2026: much smoother than the base.

Source: FRED, Federal Reserve Bank of St. Louis

$M = m \times MB$: the base quintupled, but M rose far less because the multiplier m fell as banks held excess reserves.

Excess Reserves in the Data



Excess reserves of depository institutions (millions of dollars), 2000–2020; the series was discontinued when reserve requirements were set to zero in 2020.

Source: FRED, Federal Reserve Bank of St. Louis

- ▶ Before 2008 excess reserves were essentially **zero**: $e \approx 0$.
- ▶ After the crisis they rose to over \$2.5 trillion.
- ▶ A higher e lowers the money multiplier $m = \frac{1 + c}{c + r + e}$.

Chapter summary

- ▶ Three players determine the money supply: the Fed, banks, and depositors.
- ▶ The monetary base $MB = C + R$ is the Fed's liabilities; open market purchases raise MB one-for-one, while shifts from deposits to currency leave it unchanged.
- ▶ The Fed fully controls the nonborrowed base MB_n but not borrowed reserves BR .
- ▶ In the simple model, a reserve injection creates deposits of $\Delta R/rr$ across the banking system, but a single bank can lend only its excess reserves.
- ▶ The money multiplier $m = (1 + c)/(c + r + e)$ falls when depositors hold more currency or banks hold more excess reserves – which is why QE raised the base far more than the money supply.